KEREN SUNADA MALUGURI

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SUMMARY

Graduate in Electrical & Electronics Engineering eager to leverage academic knowledge and passion for sustainable technology in an internship. Committed to contributing innovative solutions and gaining hands-on experience in EV design, development, and testing. Seeking an opportunity to apply theoretical understanding of electrical systems and emerging technologies to support the advancement of electric mobility solutions.

EDUCATION

Jawaharlal Nehru Technological Gurajada Vizianagaram (JNTGV) (2022-2025)
B.Tech. (Bachelor of Technology) in EEE. CGPA: 7.42
Damacharla Anjaneyulu Govt Polytechnic, Ongole (2019-2022)
Diploma in Electrical and Electronics Percentage: 76.67
Kendriya Vidyalaya CBSE, Ongole (2019)
SSC Percentage: 69.33

SKILLS

Programming Languages: C ,Python (basics)
Tools: Matlab Simulink, Microsoft Office (Word, PowerPoint, Excel)
Core Skills: Networks, Control Systems, DC Meachines and Transformers, Electronics
Soft Skills: Active Listener, Communication, Event Planner, Documentation and Reports preparation, Time management.

PROJECTS

Simulation of Solar Fed Buck Converter for DC Nano Applications (B.Tech Major Project)

- Designed a DC nano-grid system integrated with solar photovoltaic (PV) generation for efficient DC power distribution.
- Implemented a buck-boost DC–DC converter to obtain regulated 12V and 24V DC outputs for various DC loads.
- Developed and applied Perturb and Observe (P&O) based Maximum Power Point Tracking (MPPT) algorithm to maximize solar energy harvesting.
- Reduced power conversion losses by eliminating multiple AC–DC conversion stages.
- Simulated and validated system performance using MATLAB/Simulink under varying solar conditions.
- Ensured reliable power supply for DC loads such as mobile chargers and battery energy storage systems.
- Improved overall energy efficiency and system stability for distributed energy resource applications.

INTERNSHIP

33/11 kv Substation, Ongole, (6 Months) OCT 2021 - Mar 2022

- Learned about power distribution in Ongole substation.
- Observed various power equipment including transformers, circuit breakers, lightning arrestors, isolators, and other switchgear equipment at different voltage levels.
- Gained understanding of the functions and importance of each equipment.
- Learned about the protection scheme in Ongole for external and internal faults.

CERTIFICATIONS & Activities

- EDX Certification in Solar PV cells
- IoT and Robotics Workshop
- Python programming basics

Activity: Co-ordinated two-day national level workshop on **GENERATIVE AI** an outreach workshop of **Technex'24**, **IIT(BHU) Varanasi**.

LANGUAGES

English, Telugu, Hindi